TelConnect

Customer Churn Analysis

Predictive Modeling & Proactive Retention Strategy

Data Scientist: Lawrence Lin

Key findings & improvements

26.54%

Monthly churn rate

$359K

Projected net profit with model

+18%

Improvement vs current strategy

Proactive model-driven retention lifts net profit by +18% — from $304K to $359K

Two-intervention strategy: $18 proactive outreach + $30 reactive voucher backstop | 7,043 customers | CLV = MonthlyCharges × 6 months

26.54%

Monthly churn rate

1 in 4 customers leaves each cycle

$304K

Current net profit

Reactive-only strategy

+18%

Projected improvement

$359K with proactive strategy

Metric	Current (reactive only)	Optimized (model-driven)	Change
Net profit	$304,052	**$359,508**	**+$55,456**
Revenue recovered	$326,486	$424,515	+$98,029
Total cost	$22,434	$65,007	+$36,117
Customers proactively contacted	0	2,604	—
Customers Saved	748	962	+214

Recommendation: deploy tiered model-driven retention — $18 proactive outreach for Tier 1/2 high-risk customers; reserve $30 StreamFlix voucher for customers who call to cancel. A/B test immediately to validate the 25% proactive acceptance and the 6-month CLV horizon assumption.

Note: Optimized total cost = $46,877 proactive + $18,130 reactive = $65,007

The retention question is not 'who will churn?' — it is 'where do we spend the budget for maximum return?'

Three questions define a complete retention strategy; current approach only answers the last one

Prediction	Driver analysis	Optimization
01	02	03
Who will leave next month?	**Why are they leaving?**	**Who do we act on, and how?**
XGBoost model scores all 7,043 customers monthly. AUC 0.845. Recall 77% at optimal threshold.	Feature importance analysis (gain and SHAP) indicates contract type, tenure, and service depth as the dominant drivers for each customer.	Four-tier strategy allocates $18 outreach and $30 voucher to customers where intervention ROI is positive.

This project addresses all three — the business simulation connects model output directly to retention budget decisions.

Contract type, early tenure, and service depth account for the majority of churn risk across 7,043 customers

Overall churn rate 26.54% (1,869 churners); churners average $74/month vs $65 across all customers

7,043
Customers

26.54%
Monthly churn rate

~43%
M2M contract churn

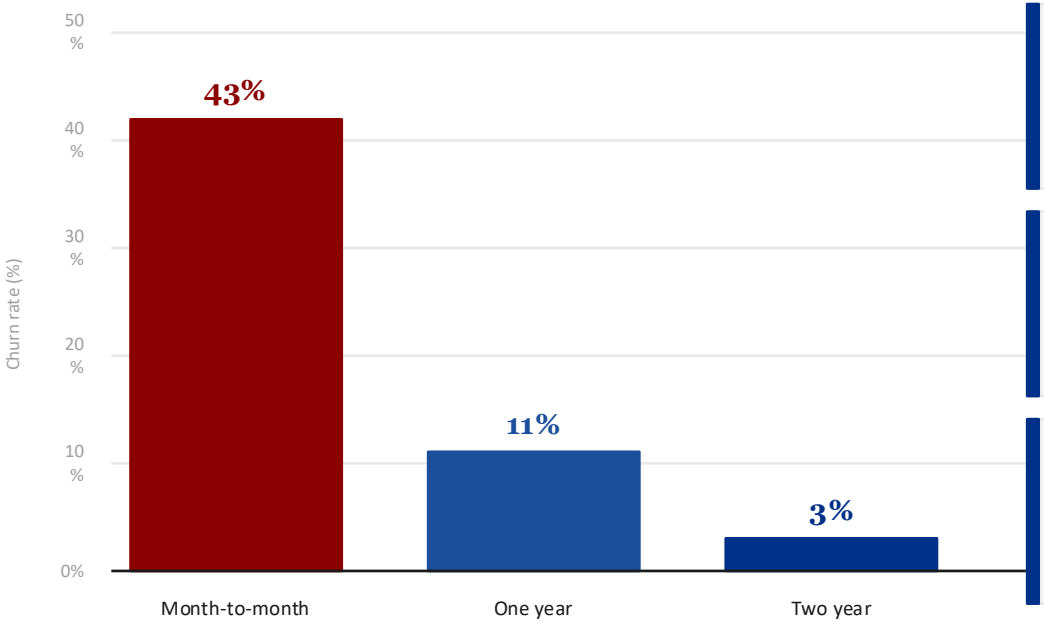
10 mo
Median churner tenure

vs 38 mo retained

1. **Contract type is dominant**
   Month-to-month: 43% churn · One-year: 11% · Two-year: 3% — a 14× differential. Contract lock-in is the strongest available retention lever.
2. **Early tenure = highest risk**
   Median churner tenure 10 months vs 38 months retained. The first 6 months is the critical window; an onboarding program directly addresses this.
3. **Fiber optic disproportionate**
   Fiber internet customers churn at materially higher rates than DSL — suggests pricing expectations or service quality issues.
4. **Service depth anchors loyalty**
   Customers without OnlineSecurity, TechSupport, or DeviceProtection churn significantly more. These services act as switching-cost anchors.
5. **Electronic check = commitment signal**
   Electronic check customers churn at 45% — nearly 3× the rate of automatic payment customers (15–17%).

Contract type is the dominant churn driver — M2M customers churn at 14× the two-year rate and account for 51% of model importance

Converting M2M customers to annual plans is the single highest-ROI retention action available



14×

Higher M2M churn vs two-year

The churn rate differential is so large that even a partial shift from M2M to annual plans produces outsized retention improvement.

50.8%

XGBoost feature importance

Contract_Month-to-month is the single dominant predictor. All other features combined account for less weight.

~$450

Avg churner CLV at risk

MonthlyCharges (average $74.40)× 6 months. Each churner prevented is $450 in retained revenue — the economic basis for the retention investment.

Note: churn rates are proportions of each contract-type group, not totals.

XGBoost achieves AUC 0.845 and 77% recall; threshold 0.56 is chosen over default 0.50 because it maximizes net profit

Three models (threshold = 0.5) evaluated via GridSearchCV · Stratified 3-fold CV · F1 scoring · class imbalance handled via weight search

Model	ROC-AUC	Recall	Precision	Notes
Logistic Regression	0.843	0.77	0.52	Interpretable baseline
Random Forest	0.838	0.80	0.52	Strong ensemble; less interpretable
XGBoost ✓ Selected	**0.845**	**0.80**	**0.52**	**Highest AUC + best net profit in profit simulation at threshold 0.56**

Why threshold 0.56, not 0.50

- Compare models at threshold = 0.5; then select the threshold that maximizes net profit for the best model.
- At 0.56: precision = 55%, recall = 77%. Accepting imperfect precision is rational: missing a churner costs ~$450 in lost CLV vs $18 for a false-positive outreach.
- The profit simulation sweeps 200 thresholds and selects 0.56 empirically as the profit-maximizing operating point.

0.845
ROC-AUC

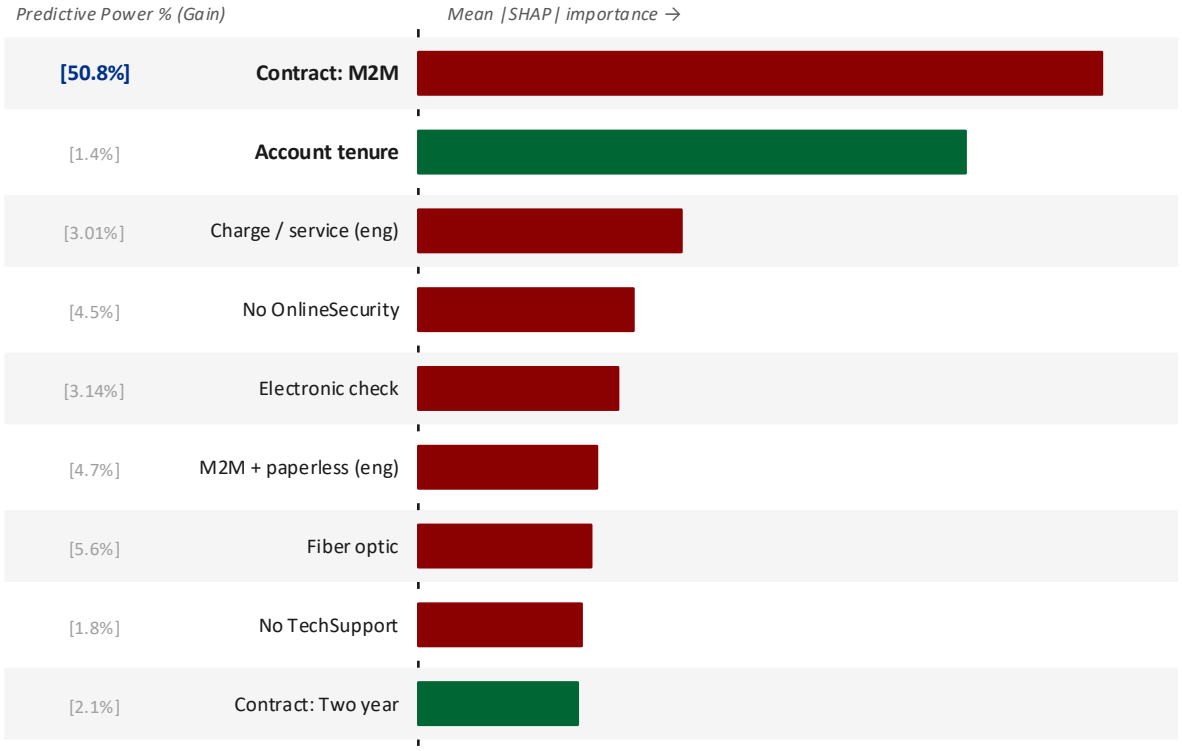
77%
Recall @ 0.56

55%
Precision @ 0.56

0.56
Optimal threshold

GridSearchCV: Logistic Regression (C, class_weight), Random Forest (max_depth, n_estimators, class_weight), XGBoost (learning_rate, max_depth, n_estimators, weight).

Contract type and tenure dominate churn prediction; service feature absence consistently increases risk



Increases churn risk

Reduces churn risk

What the data shows

① M2M contract: both metrics agree

SHAP 0.583 and gain [50.8%] are both dominant. Two independent methods, same conclusion — confidence in this finding is highest of any feature.

② Tenure: trust SHAP over gain

Gain [1.4%] drastically underestimates tenure. Continuous variables spread gain across many split thresholds. SHAP captures the true predictive weight.

③ Absent services = consistent risk

No OnlineSecurity and No TechSupport both increase churn risk. Service features act as switching-cost anchors — customers with more services have more reason to stay.

④ Two-year contract: strongest protector

Two-year contracts signal strong customer lock-in. Combined with the M2M finding: contract upgrade is the single highest-ROI retention campaign.

Mean |SHAP| · direction = sign of feature-SHAP correlation · [gain] = XGBoost built-in importance

Profit simulation across 200 thresholds selects 0.56 as the operating point — yielding $359K net profit vs $304K under the current strategy

Two-intervention model: proactive outreach ($18) + reactive voucher ($30 backstop) | scaled to full 7,043-customer base

Simulation assumptions

Parameter	Value	Source
Voucher cost	$30	Case document
Reactive acceptance	40%	Case document
CLV proxy	MC×6	Assumption*
Proactive acceptance	25%	Assumption*
Proactive outreach cost	$18	Assumption

** Key uncertainty — A/B test is the first recommended action to ensure a precise estimation.*

Strategy comparison (full customer base, n=7,043)

Metric	Current	Optimized
Net profit	$304,052	**$359,508**
Revenue	$326,486	$424,515
Proactive cost	$0	$46,877
Reactive cost	$22,434	$18,130
Total cost	$22,434	$65,007
Customers proactively contacted	0	2,604
Net profit improvement	—	**+$55,456 (+18%)**

Net profit formula

- Revenue = Proactive stream + Reactive backstop
  - Proactive stream: Proactively targeted churners × 25% × CLV
  - Reactive backstop: Churners not saved proactively × 40% × CLV
- Cost = Customers targeted × $18 + Reactive acceptances × $30
- Net profit = Revenue − Cost → maximized at threshold = 0.56

Figures scaled from held-out test set (n=1,409, scale factor 5.0×). Sensitivity analysis confirms positive ROI across proactive acceptance rates 1%–30%. MC = MonthlyCharges.
Proactive outreach costs $18 per contact regardless of acceptance (including contact and accepted incentives). Reactive costs are incurred only upon acceptance.

The proactive strategy costs $42K more but recovers $98K more revenue — net improvement of $55K with 2,604 additional customers protected

Full customer base of 7,043 customers

Current strategy — reactive only	Optimized strategy — proactive + reactive
No model-driven targeting	Model scores all 7,043 customers monthly
Voucher offered only when customer calls to cancel	2,604 Tier 1/2 customers receive $18 proactive campaign
40% of callers accept the $30 StreamFlix offer	Reactive $30 voucher reserved for call-in customers
~748 customers saved reactively per cycle	~358 proactively saved + ~604 saved reactively (+214 total incremental)
Net profit: $304,052	**Net profit: $359,508 (+18%)**
Total cost: $22,434 (reactive vouchers only)	Total cost: $65,007 (higher spend, higher return)

+$55,456 net profit · **+$98,029 revenue** · 2,604 customers proactively engaged · $42,573 additional investment

Figures scaled from test set. Optimized strategy net profit remains positive across all sensitivity scenarios tested (proactive acceptance 1%–30%).

A four-tier model assigns the right intervention to each customer by churn probability and lifetime value — avoiding blanket discounts

Model scores every customer monthly; tiers refreshed each cycle | Tier counts from test set (n=1,409), scale proportionally

Tier	Churn score	n (test)	Avg CLV	Intervention	Campaign (SHAP-matched)
Tier 1: The VIP Priority **High risk · High value**	> 0.56	387	$517	**Premium proactive outreach + reserve reactive voucher**	M2M → annual plan upgrade with lock-in incentive (higher-value discount)
Tier 2: The Efficiency Save **High risk · Lower value**	> 0.56	134	$246	**Cost-efficient proactive outreach + reserve reactive voucher**	Fiber → service quality review; no-security → free 3-month trial
Tier 3: The Nurture Watchlist **Medium risk**	0.28–0.56	285	$425	**Soft campaign, monitor**	E-check → auto-pay incentive; general loyalty touchpoint
Tier 4: The Stable Foundation Low risk	< 0.28	603	$312	**No action**	—

High-risk threshold (0.56) derived from profit simulation. High-value threshold = mean CLV of test set ($385). Tier counts scale ~5× to full population.

Four limitations require explicit disclosure — each has a concrete recommended action to resolve it before scaling the program

All assumptions stated and stress-tested in the simulation

1 Proactive acceptance rate is unknown — the largest uncertainty

The 40% reactive acceptance rate comes from customers actively trying to cancel. Proactive outreach to undecided customers will likely yield a lower rate. The simulation remains positive at acceptance rates as low as 1%, but the magnitude of the improvement is sensitive to this assumption and to proactive outreach cost.

Action: A/B test: randomize 50% of Tier 1/2 customers to outreach, 50% to control; measure acceptance rate and validate 6 months retention.

2 CLV is a rough proxy — six-month horizon understates lifetime value

MonthlyCharges × 6 ignores expected customer lifetime beyond 6 months, upgrade/downgrade probability, and cost-to-serve. True CLV could be materially higher, making the strategy even more attractive.

Action: Request survival-curve CLV estimates from finance, or tenure-weighted retention data.

3 Model identifies risk — not causation or intervention effectiveness

SHAP values show what correlates with churn, not why a specific customer will leave or which offer will retain them. A customer flagged due to fiber optic service may be dissatisfied for any of several reasons.

Action: Build an uplift model using A/B test outcomes to estimate incremental treatment effect per intervention type.

4 Single monthly snapshot limits longitudinal signal

Rolling features — charge trend over 3 months, recent service changes, support call frequency — are strong predictors unavailable in the current dataset. Including them is expected to materially improve AUC.

Action: Request rolling 6-month feature history from data engineering.

Five actions in priority sequence — the A/B test is most urgent because it validates the assumption that drives the entire ROI case

Recommended timeline: A/B test immediately; CLV data within 30 days; uplift model within 6 months

1 — Immediate

Launch A/B test

Split Tier 1/2 customers 50/50. Outreach group receives $18 proactive campaign; control does not. Measure acceptance rate and validate 6 months retention. Validates the 25% proactive acceptance assumption and provides data to refine the simulation.

2 — 30 days

Request CLV & feature data

Obtain actual CLV from Finance, and rolling 6-month features (charge trend, service changes, support calls) from Data Engineering. These strengthen model accuracy and ROI projections.

3 — 60 days

Build campaign-matching logic

Map SHAP driver profiles to specific campaign types per customer. M2M → upgrade offer; fiber → service review; no security → free trial. Increases acceptance probability vs generic outreach.

4 — 90 days

Retrain on enriched features

Once rolling features and campaign response data are available, retrain all models with longitudinal signals. Expect meaningful AUC improvement from charge trend and behavioral features.

5 — 6 months

Build uplift model

Using A/B test outcomes, estimate incremental retention effect per intervention type — not just churn probability. Moves from reactive targeting to causal budget allocation.

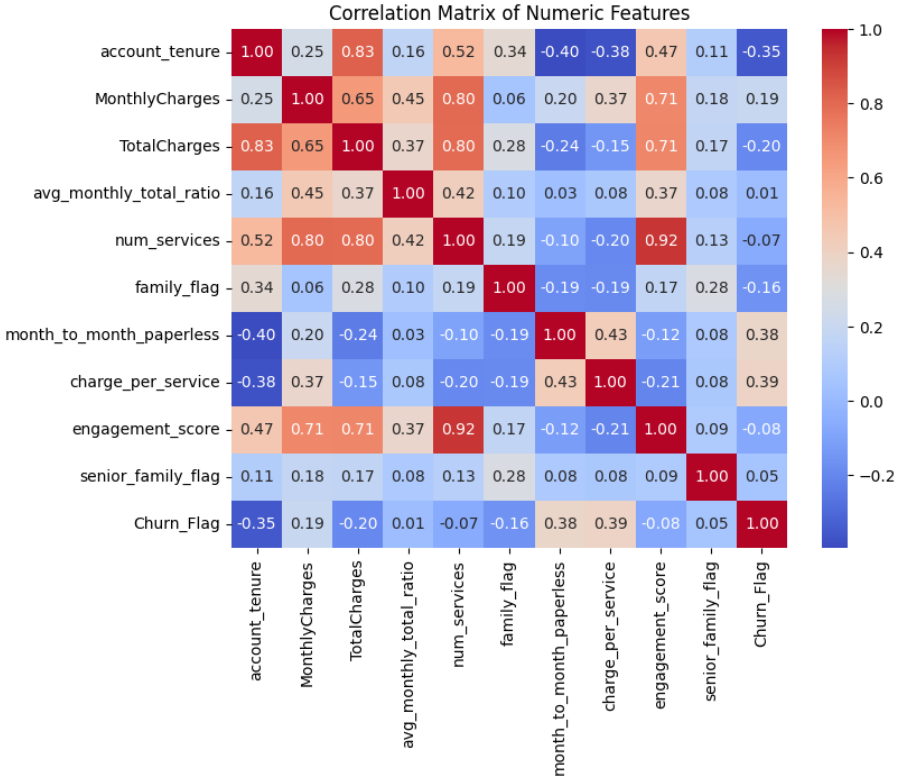
Priority determined by dependency structure: A/B test data is required before uplift modeling is possible. CLV data and feature enrichment can proceed in parallel.

Thank you for listening

Questions welcome — happy to walk through any part of the analysis in detail

Lawrence Lin · Data Scientist

A1 Correlation Matrix of Numeric Features



A2 Hyperparameter Selection with Cross-Validation

```
=== Logistic Regression Test Performance ===
              precision    recall  f1-score   support

           0       0.90      0.74      0.81      1035
           1       0.52      0.77      0.62       374

    accuracy                           0.75      1409
   macro avg       0.71      0.76      0.71      1409
weighted avg       0.80      0.75      0.76      1409

Best params: {'clf__C': 0.01, 'clf__class_weight': 'balanced'}
ROC-AUC: 0.843

=== Random Forest Test Performance ===
              precision    recall  f1-score   support

           0       0.91      0.73      0.81      1035
           1       0.52      0.80      0.63       374

    accuracy                           0.75      1409
   macro avg       0.71      0.77      0.72      1409
weighted avg       0.81      0.75      0.76      1409

Best params: {'clf__class_weight': {0: 1, 1: 6}, 'clf__max_depth': 10, 'clf__n_estimators': 100}
ROC-AUC: 0.838

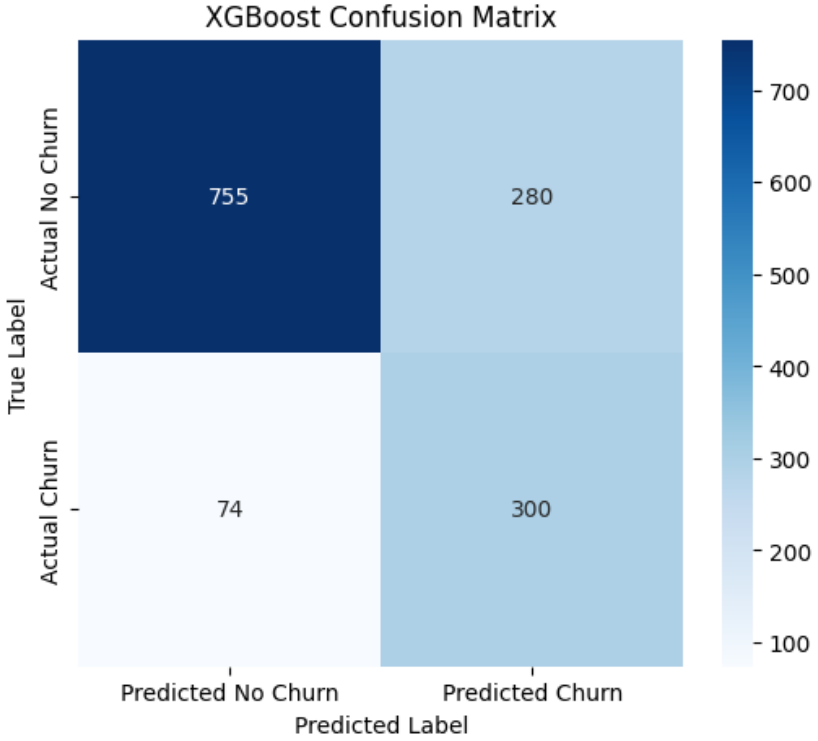
=== XGBoost Test Performance ===
              precision    recall  f1-score   support

           0       0.91      0.73      0.81      1035
           1       0.52      0.80      0.63       374

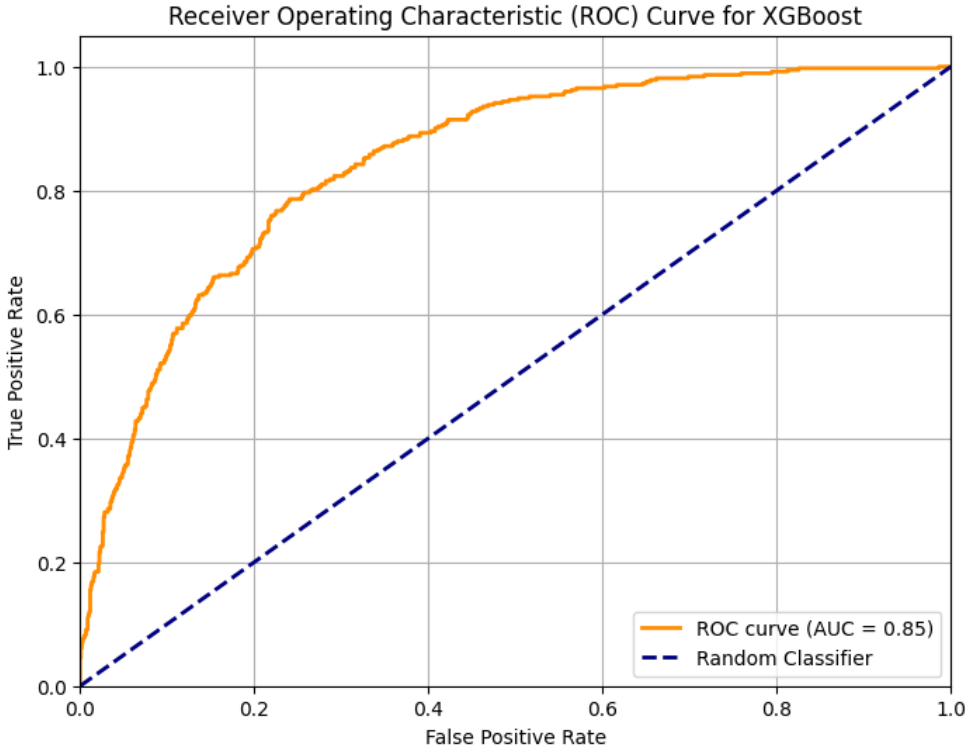
    accuracy                           0.75      1409
   macro avg       0.71      0.77      0.72      1409
weighted avg       0.81      0.75      0.76      1409

ROC-AUC: 0.845
Best params: {'clf__learning_rate': 0.1, 'clf__max_depth': 3, 'clf__n_estimators': 100, 'clf__scale_pos_weight':
np.float64(2.768561872909699)}
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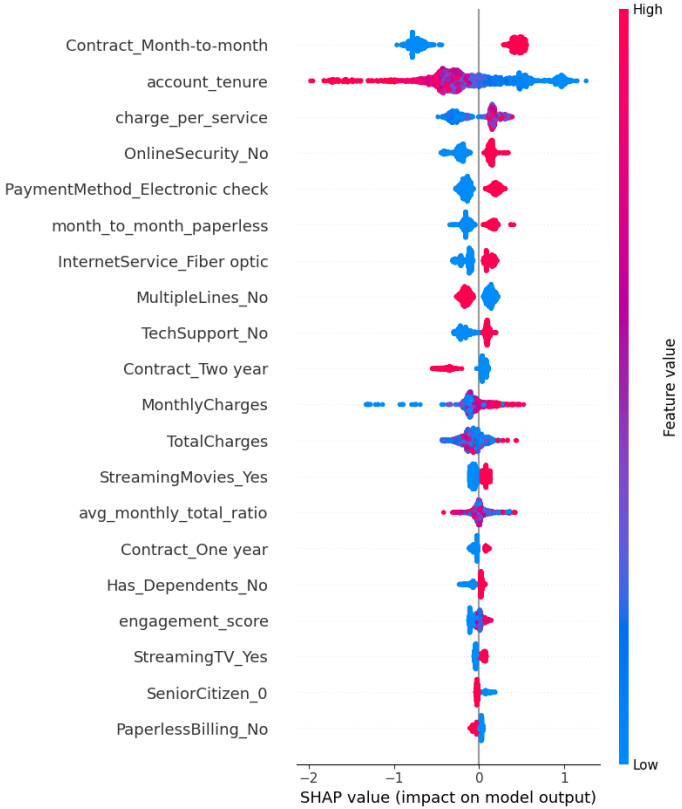
A3 XGBoost Confusion Matrix



A4 ROC Curve for XGBoost

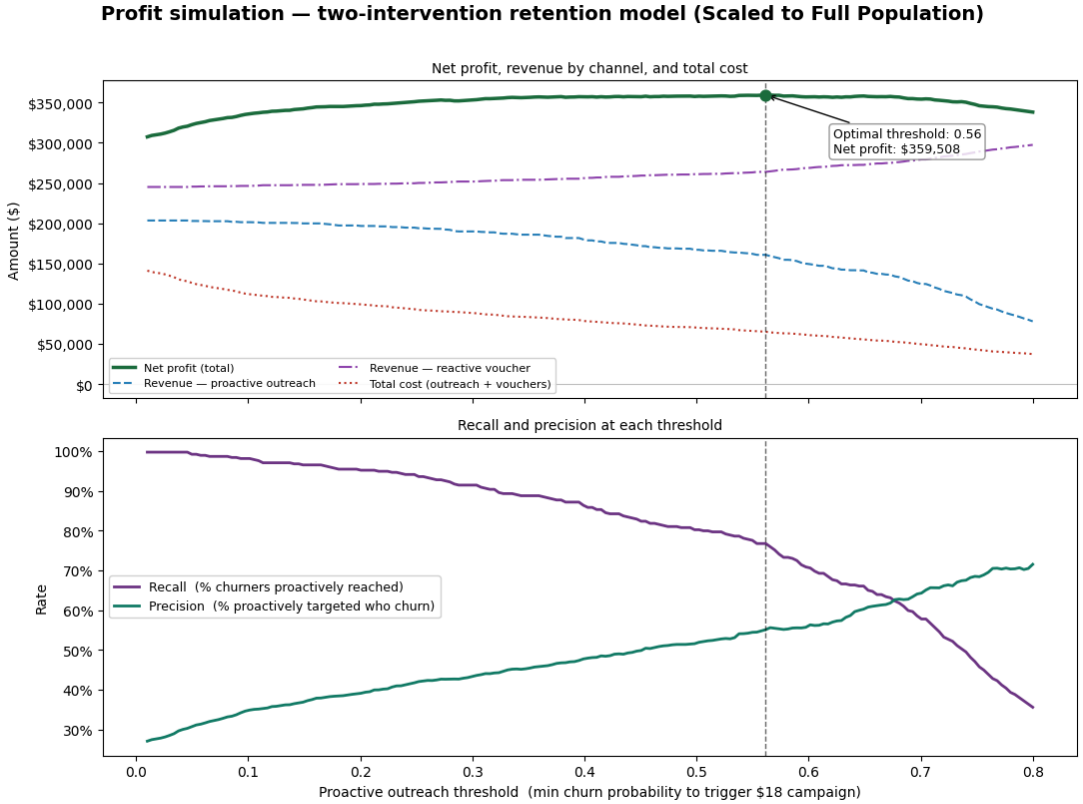


A5 SHAP Beeswarm Plot — Feature Impact on Churn Prediction

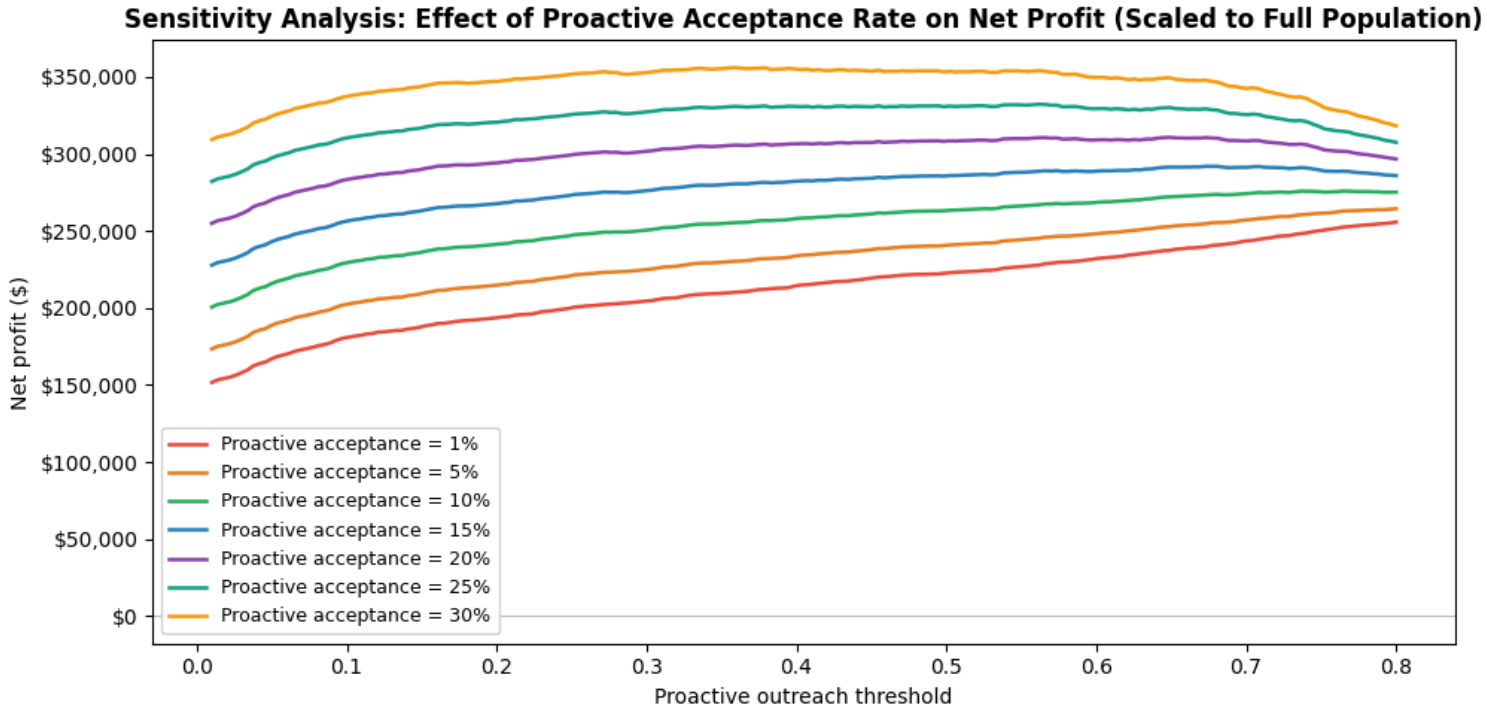



```
--- SHAP Feature Impact Summary ---
                          feature  mean_abs_shap             direction  correlation
          Contract_Month-to-month       0.582928 ▲ increases churn risk     0.994400
                   account_tenure       0.467221   ▼ reduces churn risk    -0.874502
                charge_per_service       0.225191 ▲ increases churn risk     0.681995
                  OnlineSecurity_No       0.182526 ▲ increases churn risk     0.961020
PaymentMethod_Electronic check            0.170100 ▲ increases churn risk     0.967787
       month_to_month_paperless           0.152146 ▲ increases churn risk     0.967555
   InternetService_Fiber optic            0.146745 ▲ increases churn risk     0.929048
              MultipleLines_No            0.145348   ▼ reduces churn risk    -0.966812
                TechSupport_No            0.139145 ▲ increases churn risk     0.954359
            Contract_Two year             0.135648   ▼ reduces churn risk    -0.982887
               MonthlyCharges             0.127524 ▲ increases churn risk     0.615314
                 TotalCharges             0.109137   ▼ reduces churn risk    -0.128128
```

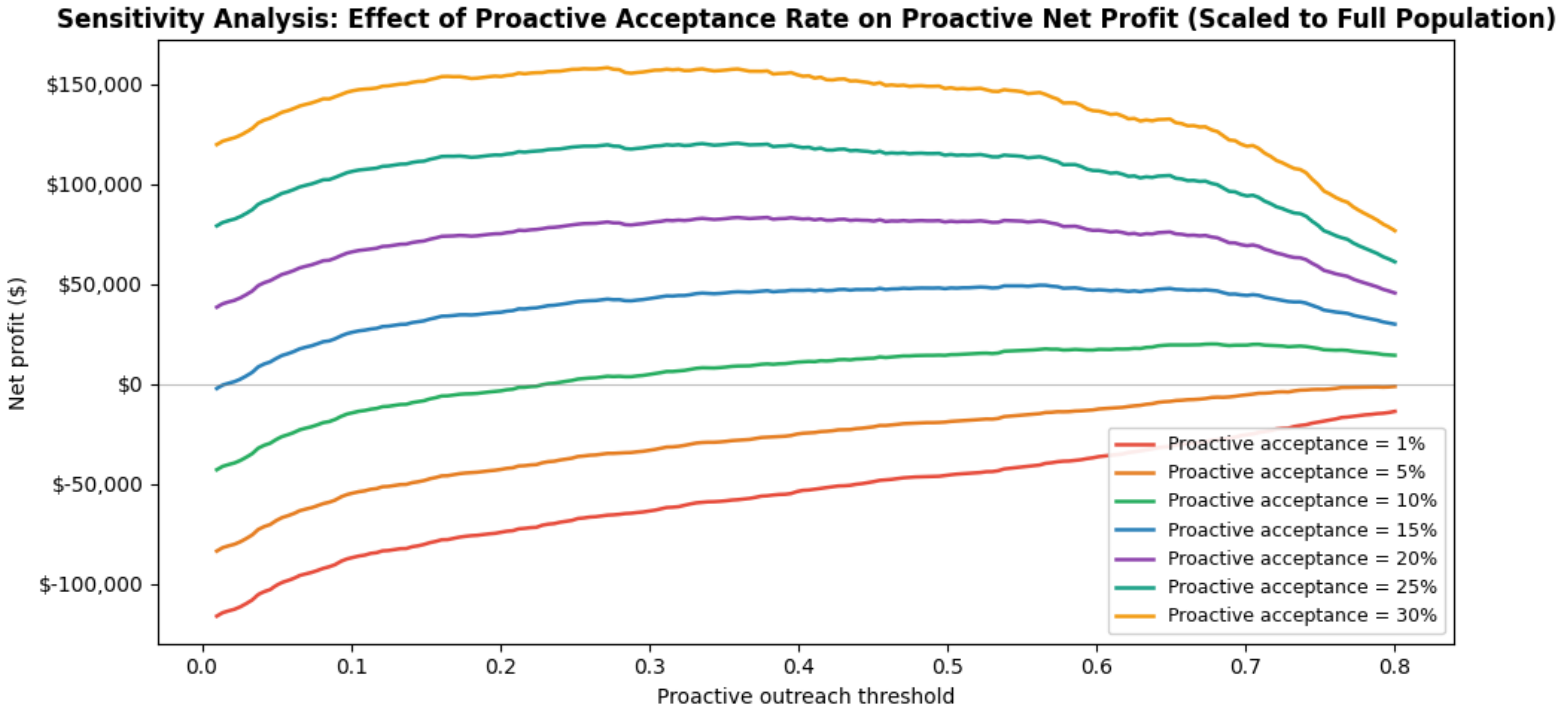
A6 Profit simulation — two-intervention retention model



A7 Sensitivity Analysis: Effect of Proactive Acceptance Rate on Net Profit



A7 Sensitivity Analysis: Effect of Proactive Acceptance Rate on Proactive Net Profit



A8 Customer segmentation: churn risk × value

